

Optimal Discounted-Cost Control of Stochastic Population Birth–Death Processes: Applications to Pest Control

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Abstract

We study stochastic optimal control problems for pest populations, with a focus on finding explicit optimal policies. Existing results often formulate optimal policies only implicitly or rely on numerical methods. Under a discounted-cost criterion, we find a closed-form optimal stationary policy for a controlled immigration model and show that it is constant with respect to the state of the process. For a controlled catastrophe model, we conjecture a policy and derive explicit closed-form bounds on the population size at which intervention begins.

1 Introduction

Agriculture is a vital component of global food supply, but unfortunately plant pests cause serious reductions in both the quantity and quality of crops [1]. Many deterministic pest-control models use systems of differential equations to predict population sizes under different controls [2, 3]. However, such models often don't incorporate the uncertainty or rare events that affect small populations. Stochastic models take into account such uncertainty and are generally formulated using stochastic differential equations or Markov chains [4].

Despite their importance, such models generally do not translate directly into decision policies. In contrast, optimal control approaches identify policies that maximize benefits while minimizing intervention costs. These strategies are often derived from an objective function using Pontryagin’s Maximum Principle [5, 6, 7]. However, they generally do not account for the stochasticity inherent in the underlying population dynamics.

Although optimal control has largely been applied to deterministic models [4], some work has considered stochastic control models. Kyriakidis [8] proves the optimality of what the author calls a control-limit policy for a continuous-time Markov decision model with catastrophe control under a long-run average-cost criterion, while Economou [9] proves threshold optimality for a controlled immigration process under both discounted and average cost criteria. However, few models provide explicit closed-form optimal policies or bounds directly in terms of model parameters. Moreover,

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much of the literature focuses on long-run average cost, whereas discounted cost is important when short-term agricultural losses and intervention costs are weighted more.

We study two controlled linear birth-death processes: one controlling pest immigration and one controlling catastrophes that reduce the pest population to zero. For the immigration model, we find a closed-form discounted-cost optimal stationary policy. For the catastrophe model, we conjecture the form of an optimal policy and derive explicit closed-form bounds on the population size at which intervention begins. Numerical results illustrate the cost implications of selecting non-optimal policies.

Organization of the paper: Section 2 presents the required theory of continuous-time Markov decision processes. Section 3 and Section 4 develop the two pest-control models, and Section 5 discusses limitations and possible extensions of this work to processes in epidemiology.

2 Preliminaries

We focus on continuous-time Markov decision processes with unbounded rates and costs, and on deterministic stationary policies that minimize the expected discounted cost. For a complete study of continuous-time Markov decision processes, we refer to [10].

The nonnegative cost Markov decision process is given by the following five-tuple.

$$\{\mathcal{S}, \mathcal{A}, (\mathcal{A}(i), i \in \mathcal{S}), q_a(j | i), c(i, a)\}.$$

The components are given as follows.

1. The state space $\mathcal{S}$ is a countable set.
2. The action space $\mathcal{A}$ is a Borel space.
3. $(\mathcal{A}(i), i \in \mathcal{S})$ is a family of nonempty measurable subsets of $\mathcal{A}$. We further let

$$\mathcal{K} := \{(i, a) : i \in \mathcal{S}, a \in \mathcal{A}(i)\},$$

be the set of all feasible state-action pairs.

4. The transition rates $q_a(j | i)$ from state i to state j satisfy $q_a(j | i) \geq 0$ for all $(i, a) \in \mathcal{K}$ and $j \neq i$. Further, we assume the transition rates are conservative, i.e.,

$$\sum_{j \in \mathcal{S}} q_a(j | i) = 0 \quad \forall (i, a) \in \mathcal{K}$$

and stable, i.e.,

$$q^*(i) := \sup_{a \in \mathcal{A}(i)} q_a(i) < \infty \quad \forall i \in \mathcal{S},$$

where $q_a(i) := -q_a(i | i)$. In addition, for each fixed $i, j \in \mathcal{S}$, the mapping $\mathcal{A}(i) \ni a \mapsto q_a(j | i)$ is measurable.

5. The cost function $c(i, a) \geq 0$ is a nonnegative measurable function on $\mathcal{K}$.

A deterministic stationary policy, or simply a stationary policy, is a function $f : \mathcal{S} \rightarrow \mathcal{A}$ such that $f(i)$ is in $\mathcal{A}(i)$ for all $i \in \mathcal{S}$. We denote the set of all stationary policies by $\mathcal{F}$.

We first impose the drift condition on the transition rates, corresponding to Assumption 2.2 of [10].

Assumption 1 (Drift condition [10]). *There exists a function $w \geq 1$ on $\mathcal{S}$ and constants $c_0 \neq 0$, $b_0 \geq 0$, and $L_0 > 0$ such that*

1. $\sum_{j \in \mathcal{S}} w(j) q_a(j | i) \leq c_0 w(i) + b_0$.
2. $q^*(i) \leq L_0 w(i)$ for all $i \in \mathcal{S}$.

Assumption 1 guarantees regularity of the minimal transition function [10].

Furthermore, for $f \in \mathcal{F}$, let $p_{i,j}^f(t)$ denote the probability that the process is in state j at time t , conditional on starting from i . Thus,

$$\sum_{j \in \mathcal{S}} p_{i,j}^f(t) = 1, \quad i \in \mathcal{S}, \quad t \geq 0.$$

Throughout, $i, j \in \mathcal{S}$ denote generic states and $i_0 \in \mathcal{S}$ an initial state.

Under $f \in \mathcal{F}$, let $I^f(t)$ denote the Markov process with $I^f(0) = i_0$ and transition rates $q_{f(k)}(j | k)$. Further, denote its law and expectation by $\mathbb{P}_{i_0}$ and $\mathbb{E}_{i_0}$, respectively. Then

$$\mathbb{E}_{i_0}[c(I^f(t), f(I^f(t)))] = \sum_{j \in \mathcal{S}} c(j, f(j)) p_{i_0,j}^f(t).$$

The expected discounted cost under policy $f \in \mathcal{F}$, starting from state $i_0 \in \mathcal{S}$, is given by

$$J_\alpha(i_0, f) = \mathbb{E}_{i_0} \left[\int_0^\infty \exp(-\alpha t) c(I^f(t), f(I^f(t))) dt \right], \quad (2.1)$$

where $\alpha > 0$ is the discount factor. The expected discounted-cost measures the expected total cost accumulated over an infinite time horizon, with future costs weighted less.

For $i \in \mathcal{S}$, define

$$J_\alpha^*(i) = \inf_{f \in \mathcal{F}} J_\alpha(i, f). \quad (2.2)$$

A policy $f^* \in \mathcal{F}$ is discounted-cost optimal if

$$J_\alpha(i, f^*) = J_\alpha^*(i) \quad \text{for all } i \in \mathcal{S}.$$

The goal in this paper is to find such discounted-cost optimal policies.

We further impose the continuity-compactness condition, from Assumption 4.12 of [10].

Assumption 2 (Continuity-compactness [10]). 1. $\mathcal{A}(i)$ is compact for each $i \in \mathcal{S}$.

2. For all $i, j \in \mathcal{S}$, the functions $a \mapsto c(i, a)$ and $a \mapsto q_a(j | i)$ are continuous on $\mathcal{A}(i)$.
3. There exists $f \in \mathcal{F}$ such that $J_\alpha(i, f) < \infty$, and $a \mapsto \sum_{j \in \mathcal{S}} J_\alpha(j, f) q_a(j | i)$ is continuous on $\mathcal{A}(i)$ for each $i \in \mathcal{S}$.

Assumption 2 ensures that the minima over the action sets are attained, allowing minimizing stationary policies to be selected [10]. Its use in numerical results is seen in Appendix B.

The following result displays the role of the discounted-cost optimality equation and corresponds to Theorem 4.6(a) of [10].

Theorem 1 (Discounted-cost optimality equation [10]). *Suppose Assumptions 1 and 2 hold. Then $J_\alpha^*(i)$ satisfies the discounted-cost optimality equation, i.e.,*

$$J_\alpha^*(i) = \min_{a \in \mathcal{A}(i)} \left\{ \frac{c(i, a)}{\alpha + q_a(i)} + \frac{1}{\alpha + q_a(i)} \sum_{j \neq i} J_\alpha^*(j) q_a(j | i) \right\}, \quad \forall i \in \mathcal{S}.$$

By Lemma 4.4 of [10] and item 3 of Assumption 2, we obtain the following result.

Proposition 1. *Suppose Assumptions 1 and 2 hold. Then the optimality equation in Theorem 1 is equivalent to*

$$\alpha J_\alpha^*(i) = \min_{a \in \mathcal{A}(i)} \left\{ c(i, a) + \sum_{j \in \mathcal{S}} J_\alpha^*(j) q_a(j | i) \right\}, \quad i \in \mathcal{S}. \quad (2.3)$$

If J_α^ satisfies one of them, then it satisfies the other.*

The next result relates the discounted-cost optimality equation to discounted-cost optimal policies and corresponds to Theorem 4.10 of [10].

Theorem 2 ([10]). *Suppose Assumptions 1 and 2 hold. Then any $f \in \mathcal{F}$ that attains the minimum in the discounted-cost optimality equation for every $i \in \mathcal{S}$ is discounted-cost optimal.*

Remark 1. *In general, the value in Equation (2.2) is taken over randomized Markov policies. Under Assumption 2, however, the minimum is attained by a deterministic stationary policy [10], which is optimal by Theorem 2. Hence, it suffices to consider deterministic stationary policies.*

Under Assumption 2, a minimizing stationary policy exists and is discounted-cost optimal by Theorem 2. To avoid requiring explicit knowledge of J_α^* , we additionally impose the following conditions from Assumptions 6.4 and 6.8 of [10].

- Assumption 3** ([10]).
1. *For every $(i, a) \in \mathcal{K}$ and some constant $M > 0$, $|c(i, a)| \leq Mw(i)$, with $w(i)$ taken from Assumption 1.*
 2. *The discount factor α satisfies $\alpha > c_0$, with c_0 taken from Assumption 1.*
 3. *There exists a nonnegative function w' on $\mathcal{S}$ and constants $c' > 0, b' \geq 0$, and $M' > 0$ such that for all $(i, a) \in \mathcal{K}$ we have*

$$q^*(i)w(i) \leq M'w'(i) \text{ and } \sum_{j \in \mathcal{S}} w'(j)q_a(j | i) \leq c'w'(i) + b'.$$

The first two items of Assumption 3 impose conditions on the behaviour of the expected discounted cost $J_\alpha(i, f)$ defined in Equation (2.1), while the final item is required for the application of Kolmogorov's forward equation (see Subsections 6.2 and 6.3 of [10]).

For w from Assumption 1 and any function u on $\mathcal{S}$, define

$$\|u\|_w := \sup_{i \in \mathcal{S}} \frac{|u(i)|}{w(i)}, \quad B_w(\mathcal{S}) := \{u : \|u\|_w < \infty\}.$$

We can now present Theorem 6.9 of [10], which provides a means for comparing discounted costs.

Theorem 3 ([10]). Suppose Assumptions 1 and 3 hold. Further, let $f \in \mathcal{F}$ be arbitrary and $V \in B_w(\mathcal{S})$. If

$$\alpha V(i) \leq c(i, f(i)) + \sum_{j \in \mathcal{S}} V(j) q_{f(i)}(j | i),$$

for all $i \in \mathcal{S}$, then $V(i) \leq J_\alpha(i, f)$ for all $i \in \mathcal{S}$.

We will illustrate in later sections that often, if $J_\alpha(\cdot, f) \in B_w(\mathcal{S})$ satisfies the discounted-cost optimality Equation (2.3), then f is discounted-cost optimal by Theorem 3.

3 Controlled Immigration of Pest Populations

The first process models trap cropping, wherein a secondary plant stand diverts target pests from the main crop and thereby prevents further immigration [11]. When $a = 1$, pests immigrate to the main crop at rate ε , and when $a = 0$, the trap crop prevents this immigration. Figure 1 illustrates the mechanism.

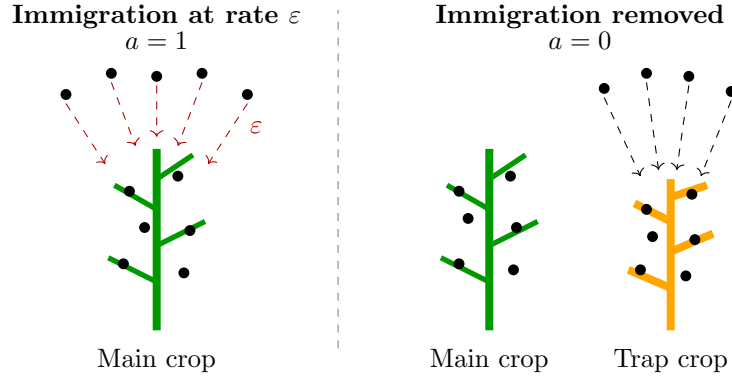


Figure 1: Illustration of immigration control. When $a = 1$, pests immigrate into the main crop at rate ε . When $a = 0$, a trap crop diverts incoming pests away from the main crop.

In the notation of Section 2, the process is presented as

$$\{\mathbb{N}_0, \{0, 1\}, (\{0, 1\}, i \in \mathbb{N}_0), q_a(j | i), iP + C(1 - a)\}, \quad (3.1)$$

with the rates given by

$$q_a(j | i) = \begin{cases} \lambda i + a\varepsilon, & j = i + 1 \\ \mu i, & j = i - 1 \\ -(\lambda + \mu)i - a\varepsilon, & j = i \\ 0, & \text{otherwise,} \end{cases} \quad (3.2)$$

for all $i, j \in \mathbb{N}_0$, with $\mathbb{N}_0 := \{0, 1, 2, \dots\}$.

Here, $\lambda, \mu, \varepsilon > 0$ are the birth, death, and immigration rates, while $P, C > 0$ are the damage cost per pest and control cost, respectively. Further, let $I^f(t)$ denote the pest population under a policy $f \in \mathcal{F}$, with $I^f(0) = i_0$.

In this section, Assumptions 1, 2, and 3 are assumed to hold, and their verification is given in Appendix A. We further impose the following parameter restrictions.

Assumption 4. *The following two conditions hold:*

1. $\lambda \neq \mu$.
2. $\alpha + \mu - \lambda > 0$.

The first condition permits a closed-form expression for the expected population size, while the second ensures convergence of the discounted-cost integrals. We proceed to present the main result of this section.

Theorem 4. *Suppose Assumptions 1, 2, 4, and 3 hold. Then a discounted-cost optimal stationary policy f^* is given by*

$$f^* = \begin{cases} f_1, & \text{if } \varepsilon P < C(\alpha + \mu - \lambda), \\ f_0, & \text{if } \varepsilon P > C(\alpha + \mu - \lambda), \end{cases}$$

where $f_1(i) = 1$ and $f_0(i) = 0$ for all $i \in \mathbb{N}_0$. If $\varepsilon P = C(\alpha + \mu - \lambda)$, both f_0 and f_1 are optimal.

Thus, this optimal policy is constant with respect to the state of the process. Immigration should be either always or never removed, with both policies achieving optimality when equality holds. The following result follows immediately from an application of Equation (7) in [12].

Proposition 2. *Suppose the first item of Assumption 4 holds. Then, given an initial state $i \in \mathbb{N}_0$,*

$$\mathbb{E}_i[I^{f_1}(t)] = i \exp((\lambda - \mu)t) + \frac{\varepsilon}{\lambda - \mu} (\exp((\lambda - \mu)t) - 1).$$

Further,

$$\mathbb{E}_i[I^{f_0}(t)] = i \exp((\lambda - \mu)t).$$

These expressions yield the below expected discounted costs.

Lemma 1. *Suppose Assumption 4 holds. Then the expected discounted cost under f_1 is given by*

$$J_\alpha(i, f_1) = P \left(i + \frac{\varepsilon}{\lambda - \mu} \right) \frac{1}{\alpha + \mu - \lambda} - \frac{P\varepsilon}{\alpha(\lambda - \mu)},$$

and the expected discounted cost under f_0 is given by

$$J_\alpha(i, f_0) = \frac{iP}{\alpha + \mu - \lambda} + \frac{C}{\alpha}.$$

Proof. By Tonelli's theorem [13, Theorem 5.28] and Proposition 2,

$$\begin{aligned} J_\alpha(i, f_1) &= P \int_0^\infty \exp(-\alpha t) \left[i \exp((\lambda - \mu)t) + \frac{\varepsilon}{\lambda - \mu} (\exp((\lambda - \mu)t) - 1) \right] dt \\ &= P \left(i + \frac{\varepsilon}{\lambda - \mu} \right) \frac{1}{\alpha + \mu - \lambda} - \frac{P\varepsilon}{\alpha(\lambda - \mu)}. \end{aligned}$$

The integral converges by Assumption 4. Similarly,

$$\begin{aligned} J_\alpha(i, f_0) &= P \int_0^\infty \exp(-\alpha t) i \exp((\lambda - \mu)t) dt + C \int_0^\infty \exp(-\alpha t) dt \\ &= \frac{iP}{\alpha + \mu - \lambda} + \frac{C}{\alpha}. \end{aligned}$$

□

We now show that f^* satisfies a particular form of the discounted-cost optimality equation.

Lemma 2. *Suppose Assumption 4 holds, and let $f \in \mathcal{F}$ take the same form as f^* in Theorem 4. Then f satisfies the alternative discounted-cost optimality Equation (2.3) for all $i \in \mathcal{S}$, i.e.,*

$$\alpha J_\alpha(i, f) = \min_{a \in \{0,1\}} \left\{ c(i, a) + \sum_{j \in \mathcal{S}} J_\alpha(j, f) q_a(j | i) \right\} \quad \forall i \in \mathcal{S}.$$

Proof. First, assume $\varepsilon P < C(\alpha + \mu - \lambda)$, so that $f = f_1$. By Lemma 1, for $i = 0$, the right-hand side of the optimality equation becomes

$$\begin{aligned} &\min_{a \in \{0,1\}} \left\{ C(1-a) + \left(P \left(1 + \frac{\varepsilon}{\lambda - \mu} \right) \frac{1}{\alpha + \mu - \lambda} - \frac{P\varepsilon}{\alpha(\lambda - \mu)} \right) \varepsilon a \right. \\ &\quad \left. - \left(P \frac{\varepsilon}{\lambda - \mu} \frac{1}{\alpha + \mu - \lambda} - \frac{P\varepsilon}{\alpha(\lambda - \mu)} \right) \varepsilon a \right\} = \min_{a \in \{0,1\}} \left\{ C(1-a) + \frac{P\varepsilon}{\alpha + \mu - \lambda} a \right\} = \frac{P\varepsilon}{\alpha + \mu - \lambda} \\ &= \alpha J_\alpha(0, f_1), \end{aligned}$$

where the minimum is attained at $a = 1$ since $\varepsilon P < C(\alpha + \mu - \lambda)$.

For $i \in \mathbb{N} := \{1, 2, 3, \dots\}$, the right-hand side becomes

$$\begin{aligned} &\min_{a \in \{0,1\}} \left\{ iP + C(1-a) + (J_\alpha(i+1, f_1) - J_\alpha(i, f_1))(\lambda i + \varepsilon a) + (J_\alpha(i-1, f_1) - J_\alpha(i, f_1))\mu i \right\} \\ &= \min_{a \in \{0,1\}} \left\{ iP + C(1-a) + \frac{P(\lambda - \mu)i}{\alpha + \mu - \lambda} + \frac{P\varepsilon}{\alpha + \mu - \lambda} a \right\} = \frac{P(\alpha i + \varepsilon)}{\alpha + \mu - \lambda} = \alpha J_\alpha(i, f_1). \end{aligned}$$

Again, the minimum is attained at $a = 1$ by $\varepsilon P < C(\alpha + \mu - \lambda)$. Hence the discounted-cost optimality equation holds for all $i \in \mathbb{N}_0$.

Now assume $\varepsilon P > C(\alpha + \mu - \lambda)$, so that $f = f_0$. For $i = 0$, Lemma 1 gives

$$\begin{aligned} &\min_{a \in \{0,1\}} \left\{ C(1-a) + \left(\frac{P}{\alpha + \mu - \lambda} + \frac{C}{\alpha} \right) \varepsilon a - \frac{C}{\alpha} \varepsilon a \right\} = \min_{a \in \{0,1\}} \left\{ C(1-a) + \frac{P\varepsilon}{\alpha + \mu - \lambda} a \right\} \\ &= C = \alpha J_\alpha(0, f_0), \end{aligned}$$

where the minimum is attained at $a = 0$ since $\varepsilon P > C(\alpha + \mu - \lambda)$.

For $i \in \mathbb{N}$, the right-hand side becomes

$$\begin{aligned} &\min_{a \in \{0,1\}} \left\{ iP + C(1-a) + (J_\alpha(i+1, f_0) - J_\alpha(i, f_0))(\lambda i + \varepsilon a) + (J_\alpha(i-1, f_0) - J_\alpha(i, f_0))\mu i \right\} \\ &= \min_{a \in \{0,1\}} \left\{ iP + C(1-a) + \frac{P}{\alpha + \mu - \lambda} (\lambda i + \varepsilon a) - \frac{P\mu i}{\alpha + \mu - \lambda} \right\} = \frac{\alpha i P}{\alpha + \mu - \lambda} + C = \alpha J_\alpha(i, f_0). \end{aligned}$$

Again, the minimum is attained at $a = 0$ by $\varepsilon P > C(\alpha + \mu - \lambda)$. Hence the discounted-cost optimality equation holds for all $i \in \mathbb{N}_0$.

Finally, if $\varepsilon P = C(\alpha + \mu - \lambda)$, then both $a = 0$ and $a = 1$ attain the minimum in the preceding calculations. Therefore, the discounted-cost optimality equation holds for either $f = f_0$ or $f = f_1$, completing the proof. $\square$

We now prove Theorem 4.

Proof. Let $f \in \mathcal{F}$ take the same form as f^* in Theorem 4. The verification that $J_\alpha(\cdot, f) \in B_w(\mathbb{N}_0)$ is given in Appendix A. By Lemma 2, for any $g \in \mathcal{F}$,

$$\alpha J_\alpha(i, f) = \min_{a \in \{0,1\}} \left\{ c(i, a) + \sum_{j \in \mathcal{S}} J_\alpha(j, f) q_a(j \mid i) \right\} \leq c(i, g(i)) + \sum_{j \in \mathcal{S}} J_\alpha(j, f) q_{g(i)}(j \mid i).$$

Hence, Theorem 3 yields

$$J_\alpha(i, f) \leq J_\alpha(i, g)$$

for all $g \in \mathcal{F}$. Therefore,

$$J_\alpha(i, f) = \inf_{g \in \mathcal{F}} J_\alpha(i, g) = J_\alpha^*(i),$$

for all $i \in \mathbb{N}_0$. Thus, f is discounted-cost optimal. $\square$

For comparison, we introduce the policy $g_m \in \mathcal{F}$, for $m \in \mathbb{N}_0$, defined by

$$g_m(i) = \begin{cases} 1, & i < m, \\ 0, & i \geq m. \end{cases}$$

Thus, under g_m , immigration is permitted while the pest population is below m and is removed from state m onward.

Figure 2 compares the optimal policy with the policies g_m . For the parameters in Subfigure 2a, immigration is optimally always removed, whereas in Subfigure 2b it is optimally always permitted. In both cases, non-optimal choices of m can result in a substantial additional expected discounted cost which is indicated by the strong contrast in colour.

Both costs in Figure 2 are numerically approximated. Moreover, the approximation method is described in Appendix B.

4 Controlled Catastrophe of Pest Populations

The next model represents interventions, such as crop removal or pesticide treatment, that eliminate the pest population. Under control ($a = 1$), pests are detected at rate d , triggering a catastrophe that resets the population to zero, and with $a = 0$, no detection is applied and the population follows the underlying birth-death process. Figure 3 illustrates the mechanism.

In the notation of Section 2, the process is described by

$$\{\mathbb{N}_0, \{0, 1\}, (\{0, 1\}, i \in \mathbb{N}_0), q_a(j \mid i), iP + Ca\}, \quad (4.1)$$



(a) Parameters: $\lambda = 0.5, \mu = 1, \varepsilon = 1, \alpha = 1, P = 200$, and $C = 100$. Optimal policy: $f^*(i) = 0$ for every considered state i .

(b) Parameters: $\lambda = 0.4, \mu = 1.2, \varepsilon = 1, \alpha = 0.8, P = 100$, and $C = 200$. Optimal policy: $f^*(i) = 1$ for every considered state i .

Figure 2: Contour plots of the additional expected discounted cost under the policy g_m relative to the optimal policy. The horizontal axis is the initial state i_0 , the vertical axis is the threshold m , and the color bar gives the numerically approximated cost difference. Source code is available in [14].

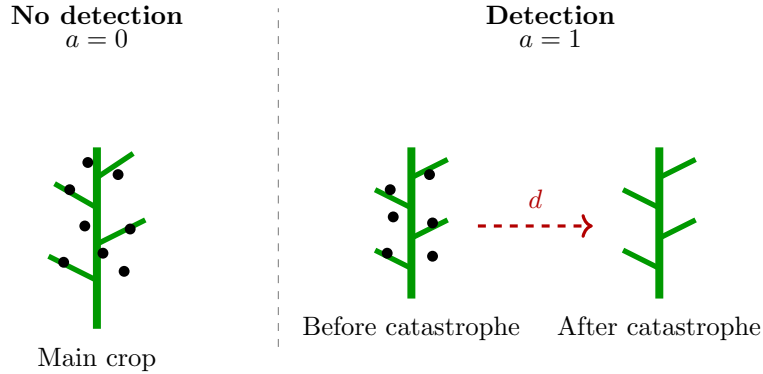


Figure 3: Illustration of the controlled catastrophe mechanism. When $a = 0$, no detection is applied. When $a = 1$, detection occurs at rate d .

with the rates given by

$$q_a(j | i) = \begin{cases} \lambda i, & \text{if } j = i + 1, \\ \mu i, & \text{if } j = i - 1, i \geq 2, \\ \mu + da, & \text{if } j = 0, i = 1, \\ da, & \text{if } j = 0, i \geq 2, \\ -(\lambda i + \mu i + da), & \text{if } j = i, i \geq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (4.2)$$

As before, $\lambda, \mu > 0$ are the birth and death rates and $P, C > 0$ are the damage and control costs; $d > 0$ is the detection rate. Again, $I(t)$ denotes the pest population, with $I(0) = i_0$.

Remark 2. Under f_0 , the catastrophe and immigration models have identical transition rates. Hence the f_0 result in Proposition 2 also applies here.

In this section, Assumptions 1 and 2 are assumed to hold, and their verification is given in Appendix A. As before, we make use of Assumption 4. We proceed to conjecture a form of an optimal policy.

Conjecture 1. A discounted-cost optimal policy for the MDP given by Equation (4.1) has the form

$$f^*(i) = \begin{cases} 0, & i < m, \\ 1, & i \geq m, \end{cases} \quad \text{for } i \in \mathbb{N}_0, \text{ and some well-chosen } m \in \mathbb{N}.$$

Conjecture 1 states that an optimal discounted-cost policy applies detection from the threshold m onward. The state dependence of this policy makes the corresponding discounted cost more challenging to evaluate. We now present the main result of this section.

Theorem 5. Suppose Assumptions 1, 2, and 4 hold. If Conjecture 1 holds, then the following inequality holds:

$$\max \left\{ \left\lceil \frac{C(\alpha + \mu - \lambda)}{dP} \right\rceil, \left\lceil \frac{C(\alpha + \mu - \lambda + d)(\alpha - d)}{\alpha dP} \right\rceil \right\} \leq m \leq \left\lfloor \frac{C(\alpha + \mu - \lambda + d)}{dP} \right\rfloor + 1,$$

where m is given by Conjecture 1.

Remark 3. In Theorem 5, when the lower and upper bounds on m coincide, the value for m is given by this bound.

Similarly to Section 3, we use f_1 and f_0 for the always-detect and never-detect policies, respectively (see Theorem 4). The next result gives an inequality for the expected number of pests under general policies.

Proposition 3. Let $f \in \mathcal{F}$ be an arbitrary admissible policy. Then, for any $i \in \mathbb{N}_0$ and $t \geq 0$,

$$\mathbb{E}_i[I^{f_1}(t)] \leq \mathbb{E}_i[I^f(t)] \leq \mathbb{E}_i[I^{f_0}(t)].$$

Proof. For $t \geq 0$, consider the stochastic processes $I^{f_1}(t)$, $I^f(t)$, and $I^{f_0}(t)$ under f_1 , f , and f_0 , respectively. We construct these processes on the same probability space such that $I^{f_1}(t) \leq I^f(t) \leq I^{f_0}(t)$ for any $t \geq 0$ almost surely.

At $t = 0$, $I^{f_1}(0) = I^f(0) = I^{f_0}(0) = i$.

If $i = 0$, then $I^{f_1}(t) = I^f(t) = I^{f_0}(t) = 0$ for all $t \geq 0$, and the required inequality follows.

Now assume that $i \geq 1$. Let $(E_i^{1,k}, E_i^{2,k}, E_i^{3,k})_{i,k \in \mathbb{N}}$ be independent random variables with distribution $\text{Exp}(1)$, and define

$$T_i^{k,\lambda} = \frac{E_i^{1,k}}{\lambda i}, \quad T_i^{k,\mu} = \frac{E_i^{2,k}}{\mu i}, \quad T_i^{k,d} = \frac{E_i^{3,k}}{d},$$

as potential waiting times for a birth, a death, and a catastrophe, respectively. Thus,

$$T_i^{k,\lambda} \sim \text{Exp}(\lambda i), \quad T_i^{k,\mu} \sim \text{Exp}(\mu i), \quad T_i^{k,d} \sim \text{Exp}(d).$$

We first couple the three processes while they are in the same state i . Their first exit times from i are

$$\begin{aligned} \tau_i^{f_1} &= \min \{T_i^{1,\lambda}, T_i^{1,\mu}, T_i^{1,d}\}, \\ \tau_i^f &= \begin{cases} \min \{T_i^{1,\lambda}, T_i^{1,\mu}, T_i^{1,d}\}, & \text{if } f(i) = 1, \\ \min \{T_i^{1,\lambda}, T_i^{1,\mu}\}, & \text{if } f(i) = 0, \end{cases} \\ \tau_i^{f_0} &= \min \{T_i^{1,\lambda}, T_i^{1,\mu}\}. \end{aligned}$$

In particular, $\tau_i^{f_1} \sim \text{Exp}(\lambda i + \mu i + d)$ and $\tau_i^{f_0} \sim \text{Exp}(\lambda i + \mu i)$.

Suppose first that $T_i^{1,\lambda}$ or $T_i^{1,\mu}$ is the minimum of the three waiting times. Then all three processes make the same transition and move to the same new state $j \in \mathbb{N}_0$. We then repeat the same construction from j .

Suppose now that $T_i^{1,d}$ is the minimum. If $f(i) = 1$, then $\tau_i^{f_1} = \tau_i^f = T_i^{1,d} < \tau_i^{f_0}$, and

$$I^{f_1}(T_i^{1,d}) = I^f(T_i^{1,d}) = 0 \leq I^{f_0}(T_i^{1,d}).$$

Since 0 is absorbing, we deduce for $t \geq T_i^{1,d}$,

$$I^{f_1}(t) = I^f(t) = 0 \leq I^{f_0}(t).$$

If $f(i) = 0$, then $\tau_i^{f_1} = T_i^{1,d} < \tau_i^f = \tau_i^{f_0}$, and

$$I^{f_1}(T_i^{1,d}) = 0 < I^f(T_i^{1,d}) = I^{f_0}(T_i^{1,d}) = i.$$

From this time onward, I^{f_1} remains at 0. We continue to couple I^f and I^{f_0} using the same birth and death clocks. As long as $f(j) = 0$ at their common state j , they make exactly the same transitions. If they reach a state j such that $f(j) = 1$, a catastrophe may occur for I^f ; in that case I^f jumps to the absorbing state 0, while I^{f_0} may not. Hence their order is preserved.

Iterating this construction yields

$$I^{f_1}(t) \leq I^f(t) \leq I^{f_0}(t), \quad t \geq 0,$$

almost surely. The required inequality follows. $\square$

Remark 4. From Proposition 3, we can infer that $\mathbb{E}_i[I^{f_1}(t)] \leq \mathbb{E}_i[I^{f_0}(t)]$ for all $i \in \mathbb{N}$ and $t \in [0, \infty)$. Moreover, by Equation (7) in [12], we have $\mathbb{E}_i[I^{f_0}(t)] < \infty$, and hence, $\mathbb{E}_i[I^{f_1}(t)] < \infty$.

The following proposition provides expressions for the expected number of pests under f_1 and f_0 .

Proposition 4. Suppose the first item of Assumption 4 holds. Then, under the initial condition $I^{f_0}(0) = I^{f_1}(0) = i \in \mathbb{N}_0$,

$$\mathbb{E}_i[I^{f_1}(t)] = i \exp((\lambda - \mu - d)t).$$

Further,

$$\mathbb{E}_i[I^{f_0}(t)] = i \exp((\lambda - \mu)t).$$

Proof. For the first claim, note that $\mathbb{E}_i[I^{f_1}] < \infty$ by Remark 4. We then define the probability generating function (PGF) as

$$G_i(\theta, t) := \mathbb{E}_i[\theta^{I^{f_1}(t)}] = \sum_{j=0}^{\infty} p_{i,j}^{f_1}(t) \theta^j,$$

where $\theta \in [0, 1]$ and $I(0) = i \in \mathbb{N}_0$.

Since the transition rates are stable and conservative, the minimal transition function satisfies the Kolmogorov equations. Moreover, under Assumption 1, it is regular (see Section 2.2 of [10] and Section 1.2 of [15]). For $|\theta| < 1$,

$$\sum_{j=0}^{\infty} p_{i,j}^{f_1}(t) |\theta|^j \leq \sum_{j=0}^{\infty} p_{i,j}^{f_1}(t) = 1,$$

so that the PGF is absolutely convergent. Furthermore, it is k -times differentiable with respect to θ for every $k \in \mathbb{N}_0$ and θ satisfying $|\theta| < 1$. In particular,

$$\lim_{\theta \rightarrow 1^-} \partial_{\theta} G_i(\theta, t) = \partial_{\theta} G_i(1, t) = \mathbb{E}_i[I^{f_1}(t)];$$

see Section 5.1 and Theorems (18) and (20) of [16].

The Kolmogorov forward equations give

$$\begin{aligned} \frac{d}{dt} p_{i,j}^{f_1}(t) &= \sum_{k=0}^{\infty} p_{i,k}^{f_1}(t) q_{f_1(k)}(j | k) \\ &= \begin{cases} \mu p_{i,1}^{f_1}(t) + d(1 - p_{i,0}^{f_1}(t)), & j = 0, \\ \lambda(j-1)p_{i,j-1}^{f_1}(t) - (\lambda j + \mu j + d)p_{i,j}^{f_1}(t) + \mu(j+1)p_{i,j+1}^{f_1}(t), & j \geq 1. \end{cases} \end{aligned}$$

Hence,

$$\begin{aligned} \partial_t G_i(\theta, t) &= \sum_{j=0}^{\infty} \frac{d}{dt} p_{i,j}^{f_1}(t) \theta^j = \mu p_{i,1}^{f_1}(t) + d(1 - p_{i,0}^{f_1}(t)) \\ &\quad + \sum_{j=1}^{\infty} [\lambda(j-1)p_{i,j-1}^{f_1}(t) - (\lambda j + \mu j + d)p_{i,j}^{f_1}(t) + \mu(j+1)p_{i,j+1}^{f_1}(t)] \theta^j. \end{aligned}$$

Differentiating with respect to θ and taking $\theta \rightarrow 1^-$ yields

$$\frac{d}{dt} \mathbb{E}_i[I^{f_1}(t)] = (\lambda - \mu - d) \mathbb{E}_i[I^{f_1}(t)].$$

Using $\mathbb{E}_i[I^{f_1}(0)] = i$, we obtain

$$\mathbb{E}_i[I^{f_1}(t)] = i \exp((\lambda - \mu - d)t).$$

The second claim follows directly from Remark 2 and Proposition 2. $\square$

The following lemma produces upper and lower bounds on the optimal discounted cost.

Lemma 3. *Suppose Assumption 4 holds. Then the optimal discounted cost satisfies*

$$\frac{iP}{\alpha + \mu - \lambda + d} \leq J_\alpha^*(i) \leq \min \left\{ \frac{iP}{\alpha + \mu - \lambda + d} + \frac{C}{\alpha}, \frac{iP}{\alpha + \mu - \lambda} \right\},$$

for all $i \in \mathbb{N}_0$.

Proof. For the upper bound,

$$J_\alpha^*(i) \leq J_\alpha(i, f_r), \quad r \in \{0, 1\}.$$

By Tonelli's theorem [13, Theorem 5.28] and Proposition 4,

$$J_\alpha(i, f_1) = \frac{iP}{\alpha + \mu - \lambda + d} + \frac{C}{\alpha}, \quad J_\alpha(i, f_0) = \frac{iP}{\alpha + \mu - \lambda}.$$

This gives the upper bound.

For the lower bound, let f^* be optimal. Since the control cost is nonnegative,

$$J_\alpha^*(i) \geq P \int_0^\infty \exp(-\alpha t) \mathbb{E}_i[I^{f^*}(t)] dt \geq \frac{iP}{\alpha + \mu - \lambda + d},$$

where the final inequality follows from Proposition 3. $\square$

Remark 5. *Lemma 3 evaluated at $i = 0$ implies $J_\alpha^*(0) = 0$.*

We have now completed the necessary prerequisites to prove Theorem 5.

Proof. Under Conjecture 1, $f^*(0) = 0$. For $i \geq 1$, the optimality equation shows that $a = 1$ is optimal whenever

$$[c(i, 0) + \sum_j J_\alpha^*(j) q_0(j | i)] - [c(i, 1) + \sum_j J_\alpha^*(j) q_1(j | i)] \geq 0 \Leftrightarrow J_\alpha^*(i) \geq \frac{C}{d}.$$

Here, the implication follows from Remark 5. By Conjecture 1, $f^*(m) = 1$, so $J_\alpha^*(m) \geq \frac{C}{d}$. Applying the two upper bounds from Lemma 3 gives

$$\begin{aligned} \frac{C}{d} &\leq J_\alpha^*(m) \leq \frac{mP}{\alpha + \mu - \lambda + d} + \frac{C}{\alpha} \Rightarrow m \geq \frac{C(\alpha + \mu - \lambda + d)(\alpha - d)}{\alpha d P} \\ \Leftrightarrow m &\geq \left\lceil \frac{C(\alpha + \mu - \lambda + d)(\alpha - d)}{\alpha d P} \right\rceil, \end{aligned}$$

and

$$\frac{C}{d} \leq J_{\alpha}^*(m) \leq \frac{mP}{\alpha + \mu - \lambda} \Rightarrow m \geq \frac{C(\alpha + \mu - \lambda)}{dP} \Leftrightarrow m \geq \left\lceil \frac{C(\alpha + \mu - \lambda)}{dP} \right\rceil.$$

Taking the maximum gives the required lower bound on m .

For the upper bound, if $m = 1$, then

$$J_{\alpha}^*(m - 1) = J_{\alpha}^*(0) = 0 \leq \frac{C}{d}.$$

If $m \geq 2$, Conjecture 1 gives $f^*(m - 1) = 0$, and hence

$$J_{\alpha}^*(m - 1) \leq \frac{C}{d}.$$

Thus, in either case,

$$J_{\alpha}^*(m - 1) \leq \frac{C}{d}.$$

Applying the lower bound from Lemma 3,

$$\begin{aligned} \frac{(m - 1)P}{\alpha + \mu - \lambda + d} &\leq J_{\alpha}^*(m - 1) \leq \frac{C}{d} \Rightarrow m - 1 \leq \frac{C(\alpha + \mu - \lambda + d)}{dP} \\ \Leftrightarrow m &\leq \left\lceil \frac{C(\alpha + \mu - \lambda + d)}{dP} \right\rceil + 1. \end{aligned}$$

Combining this with the lower bounds proves the result. $\square$

For comparison, for $m \in \mathbb{N}$, define the threshold policy $h_m \in \mathcal{F}$ by

$$h_m(i) = \begin{cases} 0, & i < m, \\ 1, & i \geq m. \end{cases}$$

Thus, under h_m , detection is not applied while the pest population is below m and is applied from state m onward.

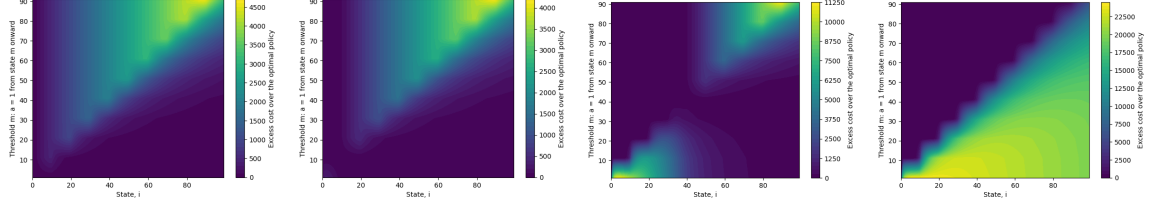
Figure 4 compares the optimal policy with the policies h_m . The parameter sets illustrate always-detect, intermediate detection, and effectively never-detect policies. In each case, non-optimal choices of m result in additional expected discounted costs.

The numerical approximations of the costs under the policies g_m seen in Figure ?? are described in Appendix B. Table 1 compares the analytical bounds on the optimal catastrophe threshold m with the numerically determined values. All numerical values lie within the predicted ranges.

5 Discussion & Further Work

5.1 Model Limitations & Further Work

The models make several subtle assumptions. For instance, their underlying population dynamics ignore seasonal effects and interactions with other species, and all parameters are assumed constant and known. In reality, controls may only reduce immigration or remove part of the population.



(a) Parameters: $\lambda = 0.5, \mu = 1, \alpha = 1, d = 1, P = 200$, and $C = 50$. Optimal threshold: $m = 1$.
(b) Parameters: $\lambda = 0.5, \mu = 1, \alpha = 1, d = 1, P = 200$, and $C = 1000$. Optimal threshold: $m = 8$.
(c) Parameters: $\lambda = 0.5, \mu = 1, \alpha = 1, d = 1, P = 1000$, and $C = 25000$. Optimal threshold: $m = 38$.
(d) Parameters: $\lambda = 0.5, \mu = 1, \alpha = 1, d = 1, P = 200$, and $C = 50000$. Optimal policy: never detect for considered states.

Figure 4: Contour plots of the additional expected discounted cost under the policy h_m relative to the optimal policy. The horizontal axis is the initial state i_0 , the vertical axis is the threshold m , and the colour bar gives the numerically approximated cost difference. Source code is available in [14].

Table 1: Analytical bounds from Theorem 5 and numerically computed values of m . Here $d = 1$. Source code is available in [14].

λ	μ	α	P	C	Analytical range for m	Numerical m
0.5	1.0	1.0	200	100	$1 \leq m \leq 2$	1
0.4	1.2	0.8	100	200	$4 \leq m \leq 6$	4
0.2	0.9	0.5	30	80	$4 \leq m \leq 6$	4
0.8	1.1	0.4	140	150	$1 \leq m \leq 2$	2
1.0	0.7	0.6	250	300	$1 \leq m \leq 2$	1
1.2	0.9	0.5	30	250	$2 \leq m \leq 11$	5
0.3	0.6	0.2	85	120	$1 \leq m \leq 3$	1
0.9	1.5	0.3	67.5	75	$1 \leq m \leq 3$	2
1.4	1.0	0.7	300	200	$1 \leq m \leq 1$	1
0.6	0.95	0.25	75	280	$3 \leq m \leq 6$	3

They also assume complete observation of the pest population, whereas monitoring may provide incomplete information. The cost functions assume linear pest damage and constant intervention costs. More realistic cost models likely change the optimal policy. Finally, numerical evidence supports the conjectured policy for the catastrophe model, but proving the optimality of this policy is left for future work.

5.2 SIS Processes

We propose two extensions to SIS processes, which are commonly used to model infections such as gonorrhea and chlamydia [17, 18, 19]. In the canonical SIS model, a fixed population of size N is partitioned into susceptible ($S(t)$) and infected ($I(t)$) individuals, with $S(t) + I(t) = N$. Infected individuals recover to susceptibility and may subsequently be reinfected, and deaths are excluded. Hence the state is determined by $I(t) \in \{0, 1, \dots, N\}$. We refer the reader to [17, 20, 21] for further information.

We first control the immigration of infected individuals. One interpretation is the isolation of the population, preventing infections acquired outside of the population. Let $\mathcal{S} = \{0, 1, \dots, N\}$, and let $a \in \{0, 1\}$, where $a = 1$ permits immigration and $a = 0$ prevents it. Thus,

$$\mathcal{A}(i) = \{0, 1\}, \quad 0 \leq i < N, \quad \mathcal{A}(N) = \{0\}.$$

Define

$$\lambda(i) = \frac{\beta i(N-i)}{N}, \quad \mu(i) = \gamma i,$$

where $\beta, \gamma > 0$ are the contact and recovery rates, and $\varepsilon > 0$ is the immigration rate. The transition rates are then given by

$$q_a(j | i) = \begin{cases} \lambda(i) + a\varepsilon, & \text{for } j = i + 1, \quad 0 \leq i \leq N - 1, \\ \mu(i), & \text{for } j = i - 1, \quad 1 \leq i \leq N, \\ -(\lambda(i) + \mu(i) + a\varepsilon), & \text{for } j = i, \quad 0 \leq i \leq N, \\ 0, & \text{otherwise.} \end{cases} \quad (5.1)$$

For $i < N$, let $c(i, a) = iP + C(1 - a)$, and set $c(N, 0) = NP$, where $P, C > 0$ are the cost per infected individual and the control cost, respectively.

The state and action spaces are finite, and the assumptions of Theorem 4.18 of [10] hold, so a discounted-cost optimal stationary policy exists.

We conjecture the following form for the optimal discounted-cost policy on the basis of numerical results.

Conjecture 2. *A discounted-cost optimal policy for the controlled-immigration SIS process has the form*

$$f^*(i) = \begin{cases} 0, & \text{for } 0 \leq i < m, \\ 1, & \text{for } m \leq i \leq N - 1, \\ 0, & \text{for } i = N, \end{cases}$$

for some $m \in \{0, 1, 2, \dots, N\}$. The case $m = N$ corresponds to $f^*(i) = 0$ for all $i \in \mathcal{S}$.

Table 2: Numerically determined optimal thresholds for the controlled-immigration SIS model, with $\varepsilon = 1$ used in all cases. Source code is available in [14].

β	γ	N	α	P	C	Numerical m
5.0	4.3	100	0.30	700	100	100
5.0	1.1	100	0.20	700	200	67
3.0	1.3	100	0.10	500	300	53
5.0	4.3	100	0.40	200	400	4
3.0	2.2	100	0.45	100	500	0

Table 2 displays thresholds obtained by discounted-cost policy iteration [10, Theorem 4.18], without imposing a threshold structure on the admissible policies. Every parameter set produces the form conjectured in Conjecture 2 (see [14] for computational details).

Since the SIS birth rates are quadratic, the equation for the mean depends on the second moment and cannot be solved directly [17], so explicit expressions for the expected discounted cost are not easily found. We therefore seek the conjectured threshold through the optimality equation.

For $0 \leq i \leq N-1$, Theorem 1 and Proposition 1 imply that $a = 1$ is optimal whenever

$$\left[c(i, 0) + \sum_{j \in \mathcal{S}} J_{\alpha}^*(j) q_0(j | i) \right] - \left[c(i, 1) + \sum_{j \in \mathcal{S}} J_{\alpha}^*(j) q_1(j | i) \right] \geq 0 \Leftrightarrow J_{\alpha}^*(i+1) - J_{\alpha}^*(i) \leq \frac{C}{\varepsilon}.$$

Thus, if the increments $J_{\alpha}^*(i+1) - J_{\alpha}^*(i)$ are above C/ε for $i < m$ and below C/ε for $i \geq m$, the conjectured policy form follows. We propose that large-population approximations may be informed by the relationship between the stochastic SIS process and its deterministic counterpart [22].

We next replace immigration control by detection control, using the same state space $\mathcal{S} = \{0, 1, \dots, N\}$. Let $a = 1$ apply detection at rate $d > 0$, after which the infected population is reset to zero, and let $a = 0$ apply no detection. Thus,

$$\mathcal{A}(0) = \{0\}, \quad \mathcal{A}(i) = \{0, 1\}, \quad 1 \leq i \leq N.$$

One interpretation is complete quarantine of the infected population following detection. With $\lambda(i)$ and $\mu(i)$ defined above, the transition rates are

$$q_a(j | i) = \begin{cases} \lambda(i), & \text{for } j = i+1, \quad 1 \leq i \leq N-1, \\ \mu(i), & \text{for } j = i-1, \quad 2 \leq i \leq N, \\ \mu(1) + da, & \text{for } j = 0, \quad i = 1, \\ da, & \text{for } j = 0, \quad 2 \leq i \leq N, \\ -(\lambda(i) + \mu(i) + da), & \text{for } j = i, \quad 1 \leq i \leq N, \\ 0, & \text{otherwise.} \end{cases} \quad (5.2)$$

We take $c(i, a) = iP + Ca$, where $P, C > 0$ have the analogous infection and control-cost interpretations. As in the preceding SIS model, finiteness of the state and action spaces guarantees the existence of a discounted-cost optimal stationary policy by Theorem 4.18 of [10].

We conjecture the following form for the optimal discounted-cost policy.

A Verification of assumptions

For Section 3, take $w(i) = i + 1$. For the first item of Assumption 1,

$$\sum_{j \in \mathbb{N}_0} w(j) q_a(j | i) = (\lambda - \mu)i + a\varepsilon,$$

with no death term when $i = 0$. Hence, we take $c_0 = \lambda - \mu \neq 0$ and $b_0 = |\lambda - \mu| + \varepsilon$, where $c_0 \neq 0$ follows from Assumption 4. For the second item, $q^*(i) = (\lambda + \mu)i + \varepsilon$, so we take $L_0 = \lambda + \mu + \varepsilon$. For Assumption 2, $\mathcal{A}(i) = \{0, 1\}$ is finite and hence compact, and the relevant mappings are continuous on this finite set. The final condition follows from Lemma 1; for each fixed state, the corresponding sum contains only finitely many nonzero terms and is therefore continuous in the action.

For Assumption 3, take $M = \max\{P, C\}$. The second item follows from $\alpha + \mu - \lambda > 0$, since $c_0 = \lambda - \mu$. For the final condition, take

$$w'(i) = (i + 1)^2, \quad M' = \lambda + \mu + \varepsilon, \quad c' = 3(\lambda + \varepsilon), \quad b' = 0.$$

Finally, using $w(i) = i + 1$ and Lemma 1,

$$\begin{aligned} \frac{J_\alpha(i, f_0)}{w(i)} &= \frac{iP}{(\alpha + \mu - \lambda)(i + 1)} + \frac{C}{\alpha(i + 1)} \leq \frac{P}{\alpha + \mu - \lambda} + \frac{C}{\alpha}, \\ \frac{J_\alpha(i, f_1)}{w(i)} &\leq \frac{P}{\alpha + \mu - \lambda} + \frac{P\varepsilon}{\alpha(\alpha + \mu - \lambda)}. \end{aligned}$$

Both bounds are finite by Assumption 4. Hence, $J_\alpha(\cdot, f_0), J_\alpha(\cdot, f_1) \in B_w(\mathbb{N}_0)$.

For Section 4, retain $w(i) = i + 1$ and $c_0 = \lambda - \mu$, and take $b_0 = |\lambda - \mu|$ and $L_0 = \lambda + \mu + d$. The first two items of Assumption 2 follow as in Section 3. The final condition follows from Remark 2 and Proposition 2. For each fixed state, only finitely many transition rates are nonzero, so the corresponding sum is continuous.

B Details of the numerical experiments

The following result is from Theorem 4.14 of [10] and makes use of Assumptions 1 and 2.

Theorem 6 ([10]). *Let $V_0^*(i) = 0$ and, for $n \geq 0$,*

$$V_{n+1}^*(i) = \inf_{a \in \mathcal{A}(i)} \left\{ \frac{c(i, a)}{\alpha + q_a(i)} + \frac{1}{\alpha + q_a(i)} \sum_{j \neq i} V_n^*(j) q_a(j | i) \right\}, \quad i \in \mathcal{S}.$$

Then

$$\lim_{n \rightarrow \infty} V_n^* = J_\alpha^*.$$

Moreover, let $f_n \in \mathcal{F}$ attain the infimum, so that

$$V_{n+1}^*(i) = \frac{c(i, f_n(i))}{\alpha + q_{f_n(i)}(i)} + \frac{1}{\alpha + q_{f_n(i)}(i)} \sum_{j \neq i} V_n^*(j) q_{f_n(i)}(j | i).$$

Then any limit point of $\{f_n\}$ is a cost-optimal policy.

Since the state space is infinite, we truncate at $k \in \mathbb{N}_0$. For large k , $c(k, f(k)) = kP + C(1 - f(k)) \approx kP$, so consecutive states have a cost that differs by approximately P . We therefore use the heuristic discounted-cost boundary condition

$$V^*(k) = V^*(k-1) + \frac{P}{\alpha}.$$

This condition is used only computationally. For Figure ??, we use 5000 iterations, truncate at $k = 250$, and plot the first 100 states to reduce boundary effects.

We validate the numerical results against the analytical costs from Lemma 1. For the parameters in Subfigures 2a,2b, the maximum absolute differences between the numerical and analytical optimal costs over the first 100 states are 4.37×10^{-11} and 3.55×10^{-11} , respectively.

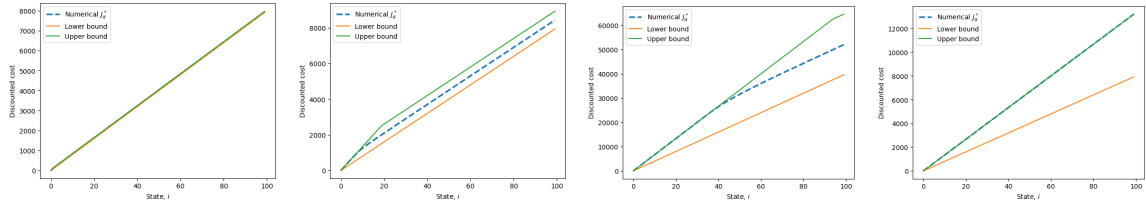


(a) Parameters and optimal policy as in Subfigure 2a.

(b) Parameters and optimal policy as in Subfigure 2b.

Figure 5: Numerical and analytical optimal discounted costs versus initial state. The numerical curve is dashed and the analytical curve solid. Source code is available in [14].

For the parameter sets in Subfigures 4a,4b,4c,4d, Figure 6 compares the numerical optimal discounted cost with the analytical bounds from Lemma 3. In every case the numerical curve remains within the bounds. Over the first 100 states, the smaller maximum absolute distances to either bound are 25.00, 500.00, 12421.82, and 4.18×10^{-11} , respectively.



(a) Parameters and optimal policy as in Subfigure 4a.

(b) Parameters and optimal policy as in Subfigure 4b.

(c) Parameters and optimal policy as in Subfigure 4c.

(d) Parameters and optimal policy as in Subfigure 4d.

Figure 6: Numerical optimal discounted cost and analytical bounds versus initial state. The numerical curve is dashed, and the upper and lower bounds are solid. Source code is available in [14].

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